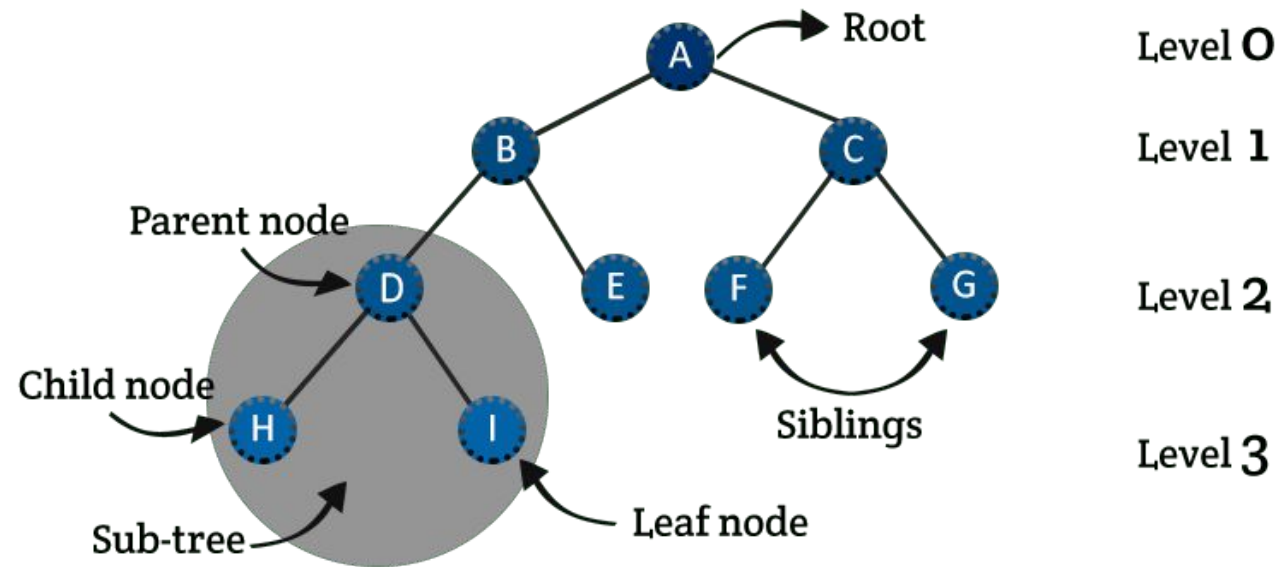


Decision Tree



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A decision tree is a specialized type of probability tree designed to aid in making decisions about a particular process. It breaks down complex problems into branches, with each branch representing a potential outcome.



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A decision tree is a specialized type of probability tree designed to aid in making decisions about a particular process. It breaks down complex problems into branches, with each branch representing a potential outcome.

- Supervised
- Classification
- Entropy
- Information Gain
- Gini Index

Decision Tree

Problem Data Set				Class
Days	Outlook	Temperature	Routine	Wear Jacket?
1	Sunny	Cold	Indoor	No
2	Sunny	Warm	Outdoor	No
3	Cloudy	Warm	Indoor	No
4	Sunny	Warm	Indoor	No
5	Cloudy	Cold	Indoor	Yes
6	Cloudy	Cold	Outdoor	Yes
7	Sunny	Cold	Outdoor	Yes

Entropy is a measure of disorder or impurity in the given dataset. In the decision tree, messy data are split based on values of the feature vector associated with each data point.

$$E(S) = \sum_{i=1}^c -p_i \log_2 p_i$$

Information Gain is the difference between the entropy of the parent node and the weighted average entropy of the child nodes.

$$\text{Gain}(T, X) = \text{Entropy}(T) - \text{Entropy}(T, X)$$

Gini Index is a powerful measure of the randomness or the impurity or entropy in the values of a dataset aims to decrease the impurities from the root nodes.

$$I_G = 1 - \sum_{j=1}^c p_j^2$$

Integer numbers

$$\begin{aligned} \log_2 16 &= \log_2 2^4 \\ &= 4 \log_2 2 \\ &= 4 \cdot 1 \\ &= 4 \end{aligned}$$

or! $\log_2 16 = \frac{\log 16}{\log 2} = \frac{\log 2^4}{\log 2} = 4$

Fraction numbers:

$$\begin{aligned} \log_2 \frac{1}{16} &= \log_2 16^{-1} \\ &= -\log_2 16 \\ &= -\log_2 2^4 \\ &= -4 \log_2 2 \\ &= -4 \cdot 1 \\ &= -4 \end{aligned}$$

Wear Jacket?		
1	YES	3 Times
2	NO	4 Times

$$E(S) = \sum_{i=1}^c -p_i \log_2 p_i$$

$E(Y)$ = Entropy Before Partition
 $E(Y|X)$ = Entropy After Partition
 Target, $E(Y) >> E(Y|X)$

Entropy Before Partition:

Entropy of Wear Jacket:

= Entropy (4, 3)
 = Entropy $(- (P_i \log_2 P_i) + (- P_i \log_2 P_i))$
 = $(-4/7 \log_2 4/7) + (-3/7 \log_2 3/7)$
 = $(-.57 \log_2 .57) + (-.43 \log_2 .43)$
 = .985 (Entropy Before Partition)

$$Gain(T, X) = Entropy(T) - Entropy(T, X)$$

$$E(S) = \sum_{i=1}^c -p_i \log_2 p_i$$

Outlook

$$E(\text{Outlook, Sunny}) = -(1/4 \log_2 1/4 + 3/4 \log_2 3/4) = .812$$

$$E(\text{Outlook, Cloudy}) = -(2/3 \log_2 2/3 + 1/3 \log_2 1/3) = .918$$

$$\begin{aligned} \text{Info Gain}(S, \text{Outlook}) &= E(S) - (4/7 * .812) - (3/7 * .918) \\ &= .985 - (4/7 * .812) - (3/7 * .918) \\ &= .127 \end{aligned}$$

Problem Data Set

Days	Outlook	Temperature	Routine	Wear Jacket?
1	Sunny	Cold	Indoor	No
2	Sunny	Warm	Outdoor	No
3	Cloudy	Warm	Indoor	No
4	Sunny	Warm	Indoor	No
5	Cloudy	Cold	Indoor	Yes
6	Cloudy	Cold	Outdoor	Yes
7	Sunny	Cold	Outdoor	Yes

$$E(S) = \sum_{i=1}^c -p_i \log_2 p_i$$

Temperature
$E(\text{Temperature, Cold}) =$ $-(1/4 \log_2 1/4 + 3/4 \log_2 3/4)$ $= .812$
$E(\text{Temperature, Warm}) =$ $-(0/3 \log_2 0/3 + 3/3 \log_2 3/3)$ $= 0.00$
$\text{Info Gain}(S, \text{Temperature}) =$ $E(S) - (4/7 * .812) - (3/7 * 0)$ $= .985 - (4/7 * .812) - (3/7 * 0)$ $= .520$

Problem Data Set

Days	Outlook	Temperature	Routine	Wear Jacket?
1	Sunny	Cold	Indoor	No
2	Sunny	Warm	Outdoor	No
3	Cloudy	Warm	Indoor	No
4	Sunny	Warm	Indoor	No
5	Cloudy	Cold	Indoor	Yes
6	Cloudy	Cold	Outdoor	Yes
7	Sunny	Cold	Outdoor	Yes

$$E(S) = \sum_{i=1}^c -p_i \log_2 p_i$$

Routine
$E(\text{Routine, Indoor}) =$ $-(1/4 \log_2 1/4 + 3/4 \log_2 3/4)$ $= .812$
$E(\text{Routine, Outdoor}) =$ $-(2/3 \log_2 2/3 + 1/3 \log_2 1/3)$ $= .918$
$\text{Info Gain}(S, \text{Routine}) =$ $E(S) - (4/7 * .812) - (3/7 * .918)$ $= .985 - (4/7 * .812) - (3/7 * .918)$ $= .127$

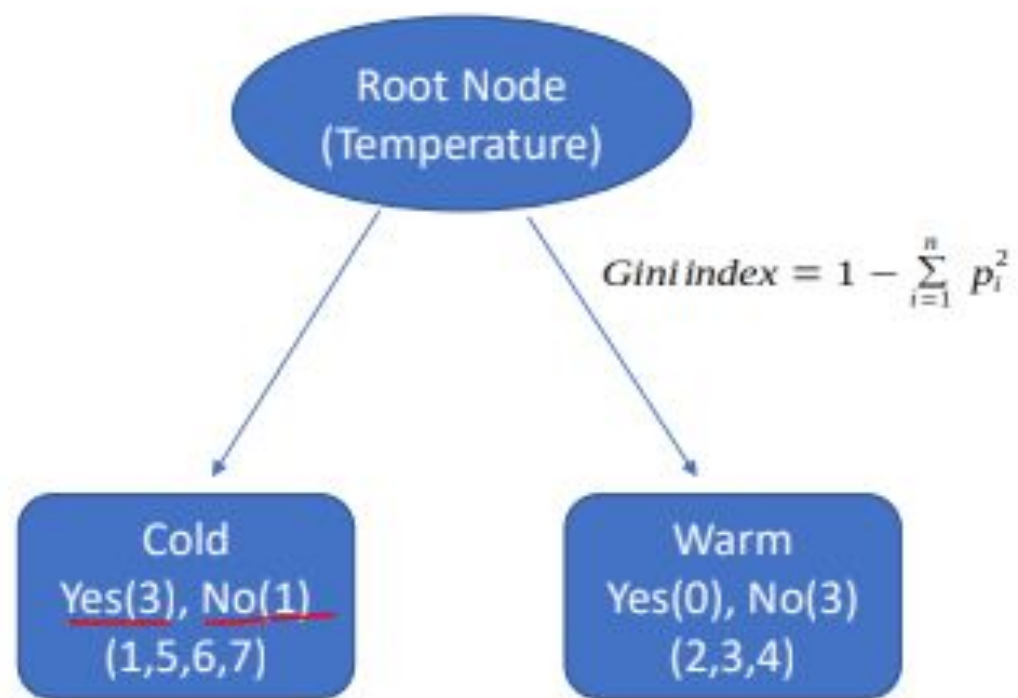
Problem Data Set

Days	Outlook	Temperature	Routine	Wear Jacket?
1	Sunny	Cold	Indoor	No
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4	Sunny	Warm	Indoor	No
5	Cloudy	Cold	Indoor	Yes
6	Cloudy	Cold	Outdoor	Yes
7	Sunny	Cold	Outdoor	Yes

$$E(S) = \sum_i -p_i \log_2 p_i$$

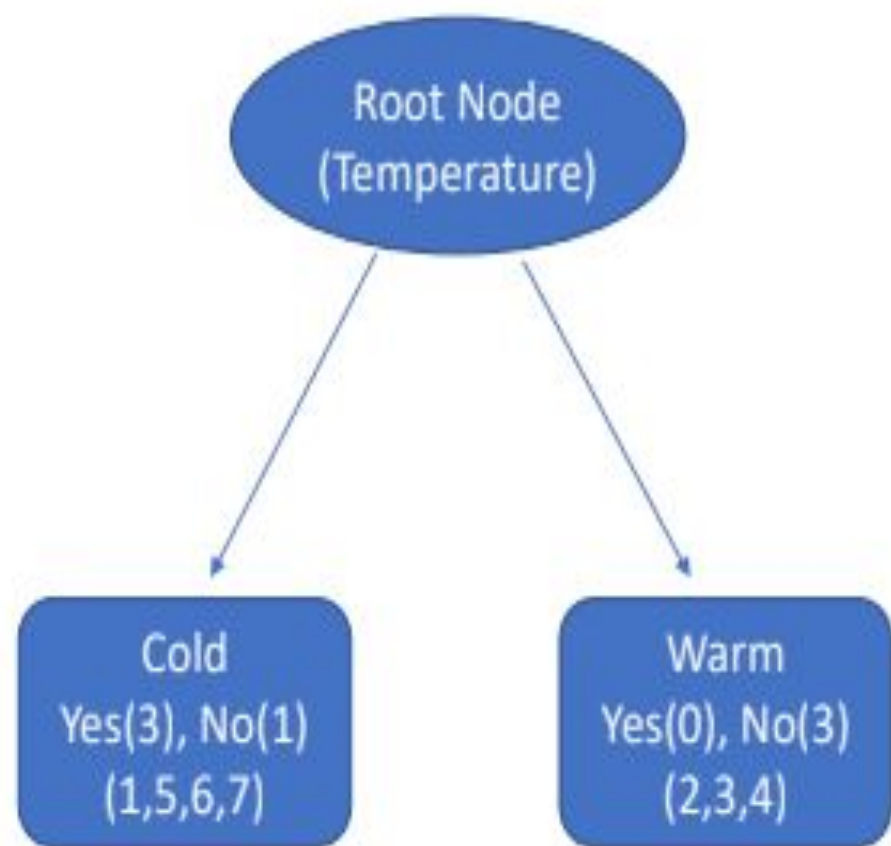
Root Node Selection Table

Outlook	Temperature	Routine
$E(\text{Outlook, Sunny}) =$ $-(1/4 \log_2 1/4 + 3/4 \log_2 3/4)$ $= .812$	$E(\text{Temperature, Cold}) =$ $-(1/4 \log_2 1/4 + 3/4 \log_2 3/4)$ $= .812$	$E(\text{Routine, Indoor}) =$ $-(1/4 \log_2 1/4 + 3/4 \log_2 3/4)$ $= .812$
$E(\text{Outlook, Cloudy}) =$ $-(2/3 \log_2 2/3 + 1/3 \log_2 1/3)$ $= .918$	$E(\text{Temperature, Warm}) =$ $-(0/3 \log_2 0/3 + 3/3 \log_2 3/3)$ $= 0.00$	$E(\text{Routine, Outdoor}) =$ $-(2/3 \log_2 2/3 + 1/3 \log_2 1/3)$ $= .918$
$\text{Info Gain}(S, \text{Outlook}) =$ $E(S) - (4/7 * .812) - (3/7 * .918)$ $= .985 - (4/7 * .812) - (3/7 * .918)$ $= .127$	$\text{Info Gain}(S, \text{Temperature}) =$ $E(S) - (4/7 * .812) - (3/7 * 0)$ $= .985 - (4/7 * .812) - (3/7 * 0)$ $= .520$	$\text{Info Gain}(S, \text{Routine}) =$ $E(S) - (4/7 * .812) - (3/7 * .918)$ $= .985 - (4/7 * .812) - (3/7 * .918)$ $= .127$



Problem Data Set

Days	Outlook	Temperature	Routine	Wear Jacket?
1	Sunny	Cold	Indoor	No
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4	Sunny	Warm	Indoor	No
5	Cloudy	Cold	Indoor	Yes
6	Cloudy	Cold	Outdoor	Yes
7	Sunny	Cold	Outdoor	Yes



Problem Data Set

Days	Outlook	Temperature	Routine	Wear Jacket?
1	Sunny	Cold	Indoor	No
2	Sunny	Warm ✖	Outdoor	No
3	Cloudy	Warm ✖	Indoor	No
4	Sunny	Warm ✖	Indoor	No
5	Cloudy	Cold	Indoor	Yes
6	Cloudy	Cold	Outdoor	Yes
7	Sunny	Cold	Outdoor	Yes

Entropy of New Subset:

$$\begin{aligned} S_2 &= \text{Entropy}(1,3) \\ &= \text{Entropy}(- (P_i \log_2 P_i) + (- P_i \log_2 P_i)) \\ &= (-1/4 \log_2 1/4) + (-3/4 \log_2 3/4) \\ &= (-.25 \log_2 .25) + (-.75 \log_2 .75) \\ &= .812 \text{ (Entropy for New Subset)} \end{aligned}$$

Problem Data Set

Days	Outlook	Temperature	Routine	Wear Jacket?
1	Sunny	Cold	Indoor	No
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3	Cloudy	Warm ✖	Indoor	No
4	Sunny	Warm ✖	Indoor	No
5	Cloudy	Cold	Indoor	Yes
6	Cloudy	Cold	Outdoor	Yes
7	Sunny	Cold	Outdoor	Yes

$$E(\text{Routine, Indoor}) =$$

$$-(1/2 \log_2 1/2 + 1/2 \log_2 1/2)$$

$$= 1$$

$$E(\text{Routine, Outdoor}) =$$

$$-(2/2 \log_2 2/2 + 0/2 \log_2 0/2)$$

$$= 0$$

$$\text{Info Gain}(S_2, \text{Routine}) =$$

$$E(S_2) - 2/4 * 1 - 2/4 * 0$$

$$= .812 - 2/4 * 1 - 2/4 * 0$$

$$= .312$$

Problem Data Set

Days	Outlook	Temperature	Routine	Wear Jacket?
1	Sunny	Cold	Indoor	No
2	Sunny	Warm ✖	Outdoor	No
3	Cloudy	Warm ✖	Indoor	No
4	Sunny	Warm ✖	Indoor	No
5	Cloudy	Cold	Indoor	Yes
6	Cloudy	Cold	Outdoor	Yes
7	Sunny	Cold	Outdoor	Yes

$$E(\text{Outlook, Sunny}) =$$

$$-(1/2 \log_2 1/2 + 1/2 \log_2 1/2)$$

$$= 1$$

$$E(\text{Outlook, Cloudy}) =$$

$$-(2/2 \log_2 2/2 + 0/2 \log_2 0/2)$$

$$= 0$$

$$\text{Info Gain}(S_2, \text{Outlook}) =$$

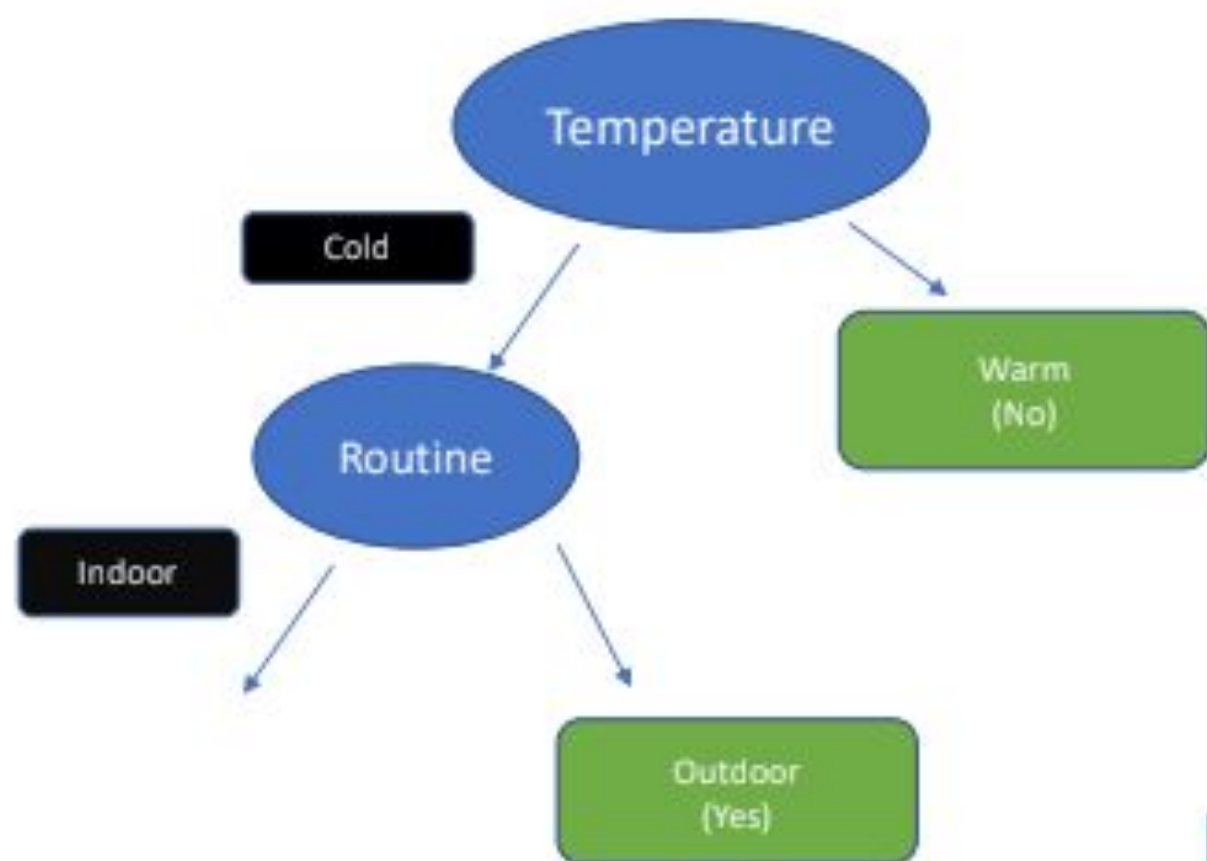
$$E(S_2) - 2/4 * 1 - 2/4 * 0$$

$$= .812 - 2/4 * 1 - 2/4 * 0$$

$$= .312$$

Problem Data Set

Days	Outlook	Temperature	Routine	Wear Jacket?
1	Sunny	Cold	Indoor	No
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5	Cloudy	Cold	Indoor	Yes
6	Cloudy	Cold	Outdoor	Yes
7	Sunny	Cold	Outdoor	Yes

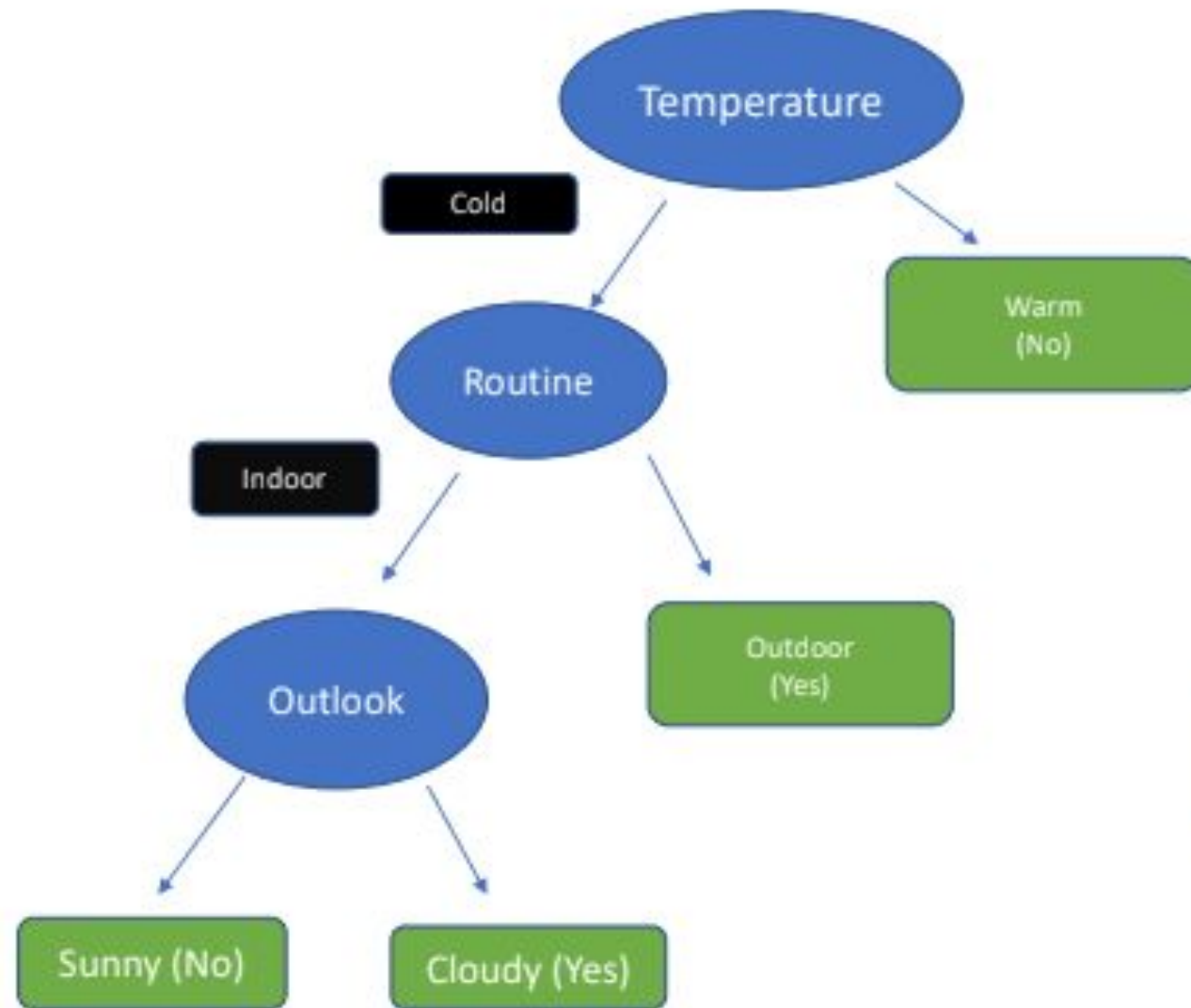


Sunny, Cold , Indoor= ??



Problem Data Set

Days	Outlook	Temperature	Routine	Wear Jacket?
1	Sunny	Cold	Indoor	No
2	Sunny	Warm ✖	Outdoor	No
3	Cloudy	Warm ✖	Indoor	No
4	Sunny	Warm ✖	Indoor	No
5	Cloudy	Cold	Indoor	Yes
6	Cloudy	Cold	Outdoor ✖	Yes
7	Sunny	Cold	Outdoor ✖	Yes



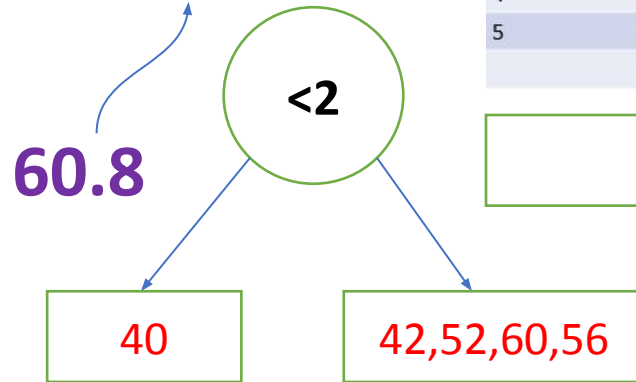
Sunny, Cold , Indoor= ??

DECISION TREE REGRESSION

Experience	Gap	Salary
2	Yes	40
2.5	Yes	42
3	No	52
4	No	60
5	Yes	56
	Avg	50

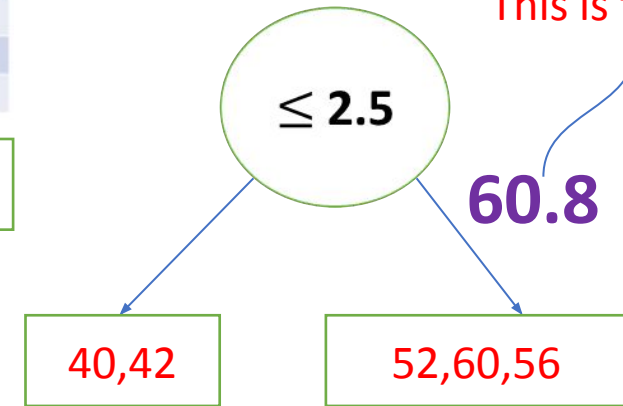
Experience	Gap	Salary
2	Yes	40
2.5	Yes	42
3	No	52
4	No	60
5	Yes	56
Avg		50

This is the root variance



40,42,52,60,56

This is the root variance



To select the best split, we use the variance reduction technique.

$$\text{Variance} = \frac{1}{n} \sum_{i=1}^M (y_i - y^{\text{pred}})^2$$

$$\begin{aligned} \text{Variance Root} &= \frac{1}{5} ((40 - 50)^2 + (42 - 50)^2 + (52 - 50)^2 + (60 - 50)^2 + (56 - 50)^2) \\ &= \frac{1}{5} (10 + 64 + 4 + 100 + 36) = 60.8 \end{aligned}$$

Variance for Child 1 1st split :

$$\text{child 1} = \frac{1}{1} (40 - 50)^2 = 100$$

$$\text{child 2} = \frac{1}{4} (64 + 4 + 100 + 36) = 51$$

Variance for Child 1 2nd split :

$$\text{child 1} = \frac{1}{2} (100 + 64) = 164/2 = 82$$

$$\text{child 2} = \frac{1}{3} (4 + 100 + 36) = \frac{140}{3} = 46.6$$

Variance Reduction:

$$\text{var}(\text{root}) - \sum w_i \text{var}(\text{child1})$$

$$= 60.8 - \left[\frac{1}{5} * 100 + \frac{4}{5} * 51 \right]$$

$$= 60.8 - [20 + 40.8]$$

$$= 0$$

Variance Reduction:

$$\text{var}(\text{root}) - \sum w_i \text{var}(\text{child1})$$

$$= 60.8 - \left[\frac{2}{5} * 82 + \frac{3}{5} * 46.6 \right]$$

$$= .304$$

